

Akash Khatri

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EDUCATION

Master of Science in Computer Science <i>University of Utah - GPA: 4.0/4.0</i>	Aug 2023 – May 2025
Bachelor of Engineering in Computer Engineering <i>University of Mumbai - GPA: 9.64/10</i>	Aug 2019 – May 2023

TECHNICAL SKILLS

Languages: Java, Python, Go, JavaScript, TypeScript, C, SQL
Frameworks & Libraries: React, Node.js, Express, FastAPI, Flask, Jinja2, LangChain
Cloud & Data Engineering: AWS, CI/CD pipelines, ETL Workflows, Data Pipelines
Databases & Vector Search: PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, LLMs, LangChain, OpenAI APIs, NLP

WORK EXPERIENCE

Software Development Engineer <i>Amazon</i>	Jun 2025 – Present
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- Architect and scale backend services and AWS data pipelines in **Java**, orchestrating production ETL workflows and linking high-throughput product ordering pipelines to centralized commerce layers serving **millions of users daily**.
- Coordinated the cross-functional release lifecycle of the **Amazon Music for Artists** application; unblocked multi-platform deployments across Web, Android, and iOS by resolving critical defects, directing QA testing in beta environments, and executing multi-channel MCMs.
- Owned operational resilience for core microservices by structuring multi-package beta pipelines and implementing self-triggering **canary stacks with active rollback protection** to safeguard production deployment runs.
- Executed data layer migrations for high-volume **AWS Glue** pipelines handling **TBs of historical data**; designed validation tooling that eliminated duplicate backfill processing routines, dropping infrastructure overhead by **\$50K+ annually**.
- Led the testing infrastructure migration of **20+ dependent packages** to a runtime-managed framework; authored the technical design doc and formulated **AWS CDK constructs** to parameterize environment setups, compute scaling (Fargate/Lambda), and artifact generation.
- Orchestrated the component-by-component rollout across isolated distributed applications by mapping Jira epics, mentoring and unblocking a junior SDE-1 throughout the integration to deliver the entire lifecycle **2 weeks ahead of schedule**.

AI/ML Research Assistant <i>Kahlert School of Computing, University of Utah</i>	Apr 2024 – May 2025
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- First-authored “Sort it Like You Mean It” (**PVLDB 2025**) and deployed **InsightSort**, a semantic analytical engine running on **GPT-4o** that maps 5–7 context-aware ranking dimensions onto target datasets, enriching tabular records with schemas completely missing from the host table.
- Engineered a dual 384-dimensional **HNSW vector-index layer** over extensive data lakes (isolating column joinability and column semantics) that replaces $O(N)$ brute-force column scans with sub-linear top-5 approximate nearest neighbor lookups to isolate valid join paths in milliseconds.
- Formulated a structural retrieval criteria fusing multi-modal utility scores with embedding-based uniqueness markers to filter out redundant relational keys, instantly synthesizing optimized **DuckDB SQL** query patches to skip manual data profiling leaks.

Graduate Teaching Assistant <i>Kahlert School of Computing, University of Utah</i>	Jan 2024 – May 2025
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- Conducted labs and evaluated core systems for Advanced Algorithms and Full-Stack Systems (**React, FastAPI, AWS**) for **180+ graduate students**; provided architectural mentorship on concurrent state software and scaling configurations.

PUBLICATIONS

- Akash Khatri**, et al. “Sort it Like You Mean It: Discovering Semantically Interesting Attribute Augmentations to Sort Tables” | *Proceedings of the VLDB Endowment (PVLDB)*, Vol. 18, No. 12, 2025.

PROJECTS

SmartBillAgent: Conversational Billing & Document Automation Platform <i>Python, Flask, Anthropic, Docker</i>	GitHub
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- Constructed an asynchronous multi-threaded **Flask** backend to process conversational text orders from Telegram webhooks, eliminating manual pen-and-paper billing workflows to scale multi-store operations to **60+ daily orders** and **\$110,000+** in gross transactional volume.
- Forged a fault-tolerant semantic parser using **Claude LLM constraints** that maps vague inputs across English, Hindi, and Sindhi into schema-validated JSON data contracts with programmatic bilingual rendering to bypass worker literacy constraints.
- Devised an on-the-fly bill-compilation pipeline using **Jinja2** templates to process raw JSON arrays into paginated, print-ready HTML and synthesize structural vector PDF receipts within seconds via cloud endpoints.

Raft Distributed Consensus Engine <i>Go, Distributed Systems, Concurrency, RPC</i>	GitHub
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- Created a production-spec **Raft consensus module** in Go featuring automated leader election, heartbeats, continuous log replication mechanisms, and persistent state machines over isolated nodes.
- Validated system core through testing harnesses mimicking dynamic network partitions and cascade node drops across **10+ simulation clusters**, asserting **100% state data correctness** under intense split-brain scenarios.